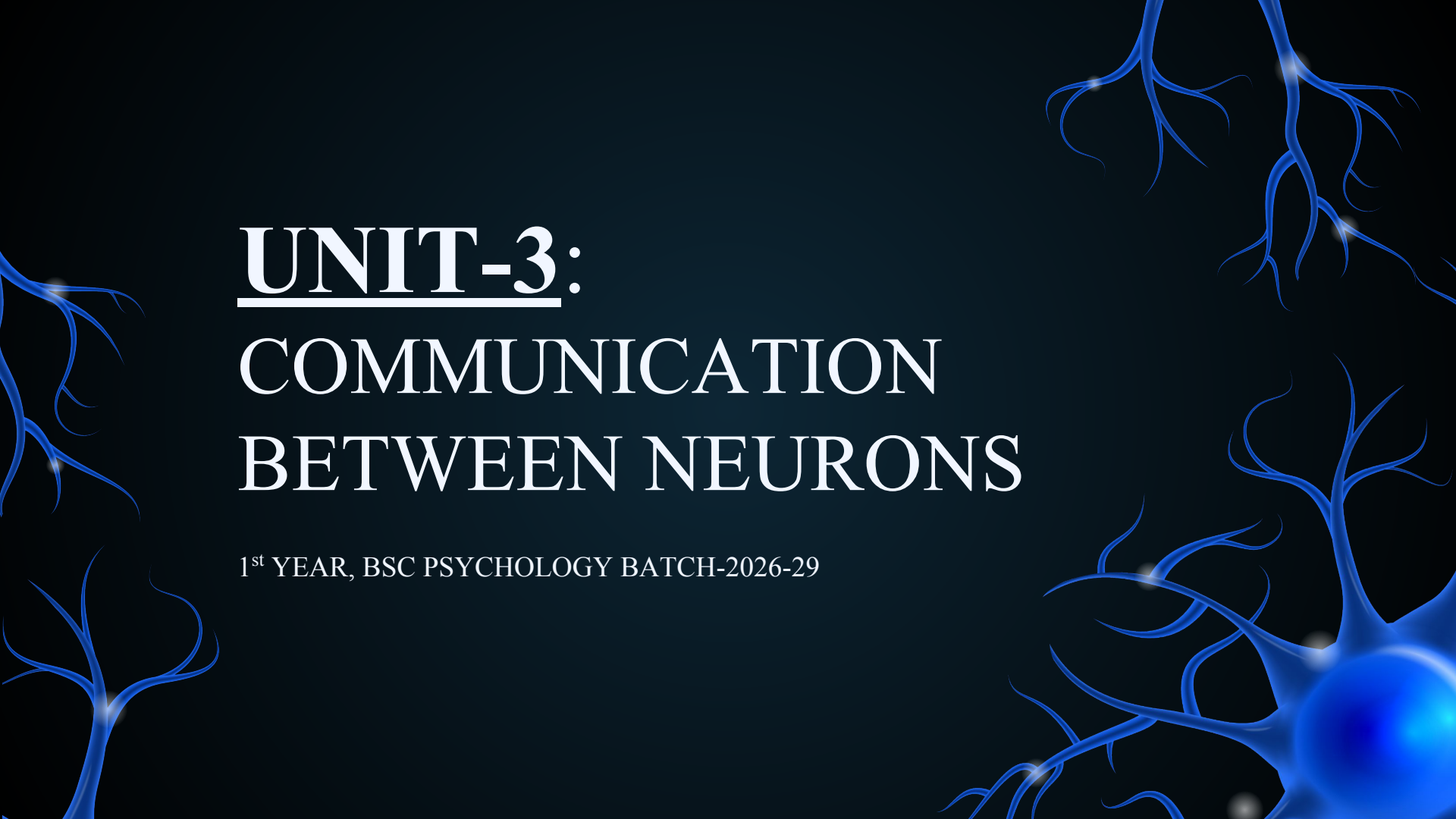


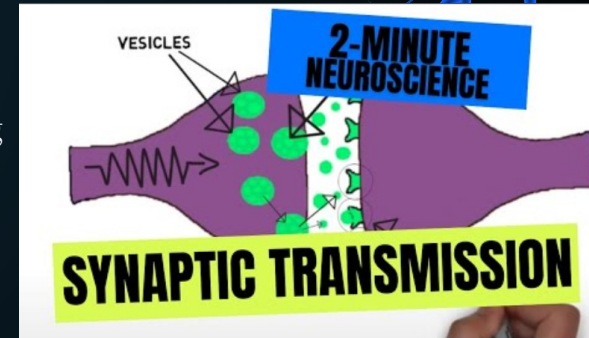
The background of the slide is a dark navy blue. It is decorated with several stylized, glowing blue neurons. These neurons have a central cell body (soma) and multiple branching processes (dendrites and axons) that extend across the frame. Some of the branching points and the cell bodies themselves have a bright, glowing blue light effect, giving them a three-dimensional, ethereal appearance. The neurons are positioned in the corners and along the sides, framing the central text.

UNIT-3: COMMUNICATION BETWEEN NEURONS

1st YEAR, BSC PSYCHOLOGY BATCH-2026-29

Introduction

- ❑ In the **late 1800s** ❑ Ramón y Cajal **anatomically** demonstrated a narrow gap separating one neuron from another.
- ❑ In **1906** ❑ Charles Scott Sherrington **physiologically** demonstrated that communication between one neuron and the next differs from communication along a single axon. He inferred a specialized gap between neurons and introduced the term **synapse** to describe it.
- ❑ **Definition of Synapse** ❑ A specialized junction between two neurons where one neuron communicates with another by releasing **neurotransmitters across the synaptic cleft**.
- ❑ Cajal and Sherrington are regarded as the great pioneers of modern neuroscience.



The Properties of Synapses

- ❑ Sherrington studied **reflexes**—automatic muscular responses to stimuli.
 - ❑ In a **leg flexion reflex**, a sensory neuron excites a second neuron, which in turn excites a motor neuron, which excites a muscle. The circuit from sensory neuron to muscle response is called a **reflex arc**.
- If one neuron is separate from another, as Cajal had demonstrated, how does the message get from one neuron to the next?
- ❑ Sherrington realized that a reflex must require communication between neurons, and therefore, measurements of reflexes might reveal some of the special properties of that communication.
 - ❑ To measure reflexes, Sherrington strapped a dog into a harness above the ground and pinched one of the dog's feet. After a fraction of a second, the dog flexed (raised) the pinched leg and extended the other legs.
 - ❑ Sherrington found the same reflexive movements after he made a cut that disconnected the spinal cord from the brain.
 - ❑ Evidently, the spinal cord controlled the flexion and extension reflexes.
 - ❑ In fact, the movements were more consistent after he separated the spinal cord from the brain.
 - ❑ Sherrington observed several **properties of reflexes suggesting special processes at the junctions between neurons:**
 - a) Reflexes are slower than conduction along an axon.
 - b) Several weak stimuli presented at slightly different times or locations produce a stronger reflex than a single stimulus does.
 - c) When one set of muscles becomes excited, a different set becomes relaxed.



The Properties of Synapses

a) Speed of a Reflex and Delayed Transmission at the Synapse

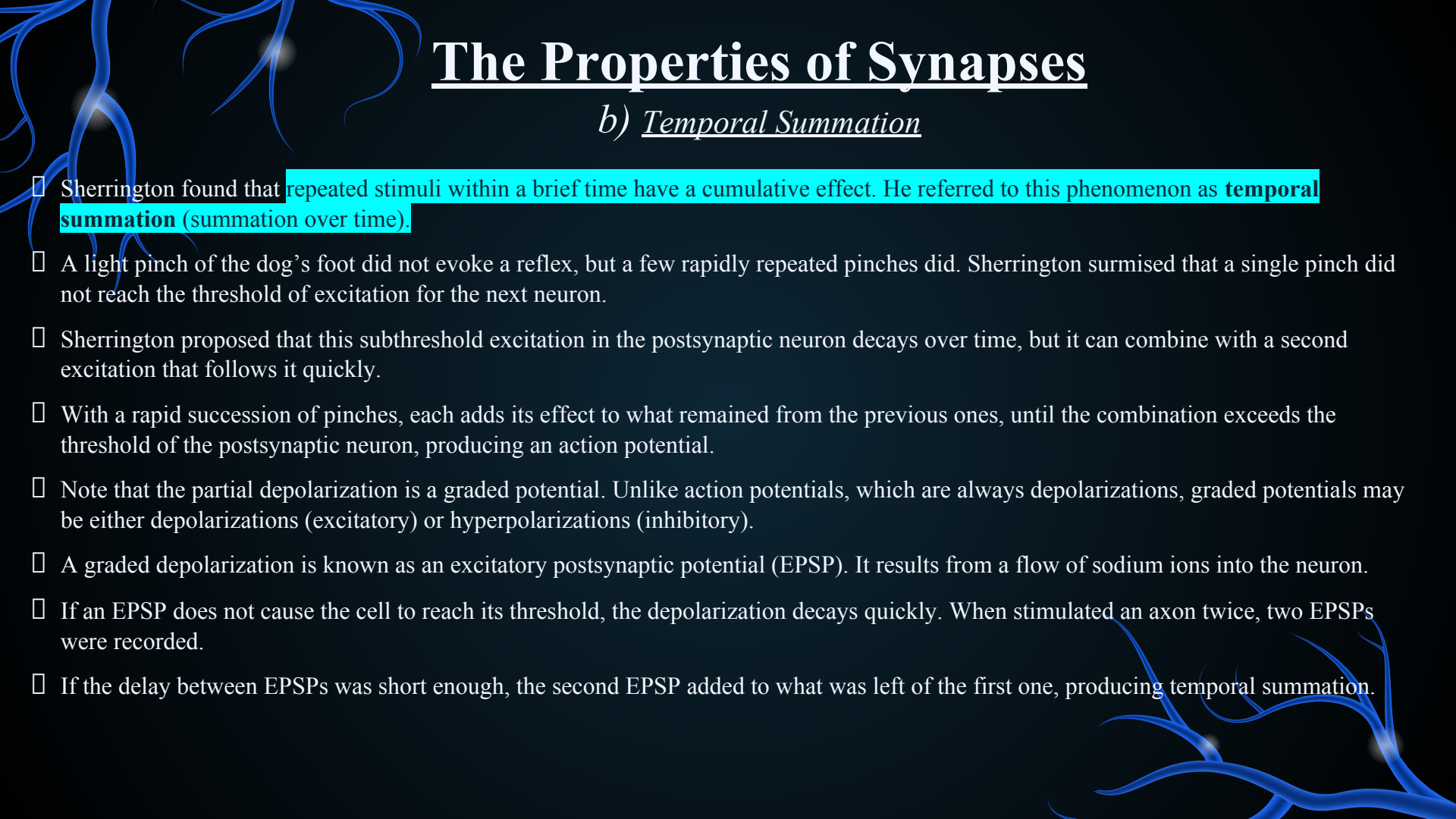
- When Sherrington pinched a dog's foot, the dog flexed that leg after a short delay.
- During that delay, an impulse had to travel up an axon from the skin receptor to the spinal cord, and then an impulse had to travel from the spinal cord back down the leg to a muscle.
- Sherrington measured the total distance that the impulse traveled from skin receptor to spinal cord to muscle and calculated the speed at which the impulse must have traveled to produce the response.
- He found that the speed of conduction through the reflex arc varied but was never more than about 15 meters per second (m/s).
- In contrast, previous research had measured action potential velocities along sensory or motor nerves at about 40 m/s.
- Sherrington concluded that some process was slowing conduction through the reflex, and he inferred that the delay must occur where one neuron communicates with another
- This idea is critical, as it established the existence of synapses. Sherrington, in fact, introduced the term synapse.



The speed of conduction along an axon is about 40 m/s.



The speed of conduction through a reflex arc is slower and more variable, sometimes 15 m/s or less. Presumably, the delay occurs at the synapse.



The Properties of Synapses

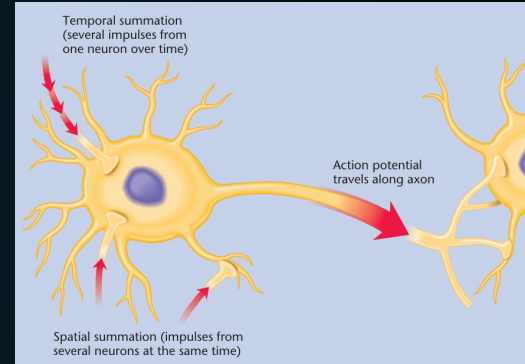
b) Temporal Summation

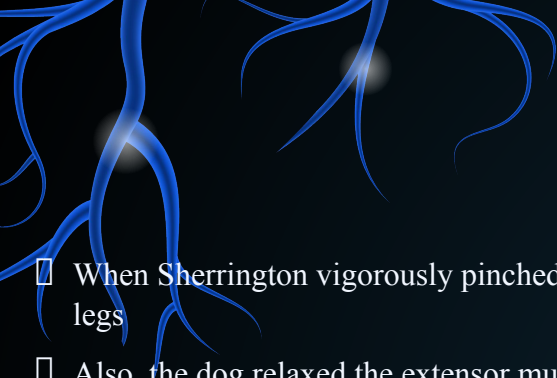
- Sherrington found that repeated stimuli within a brief time have a cumulative effect. He referred to this phenomenon as **temporal summation** (summation over time).
- A light pinch of the dog's foot did not evoke a reflex, but a few rapidly repeated pinches did. Sherrington surmised that a single pinch did not reach the threshold of excitation for the next neuron.
- Sherrington proposed that this subthreshold excitation in the postsynaptic neuron decays over time, but it can combine with a second excitation that follows it quickly.
- With a rapid succession of pinches, each adds its effect to what remained from the previous ones, until the combination exceeds the threshold of the postsynaptic neuron, producing an action potential.
- Note that the partial depolarization is a graded potential. Unlike action potentials, which are always depolarizations, graded potentials may be either depolarizations (excitatory) or hyperpolarizations (inhibitory).
- A graded depolarization is known as an excitatory postsynaptic potential (EPSP). It results from a flow of sodium ions into the neuron.
- If an EPSP does not cause the cell to reach its threshold, the depolarization decays quickly. When stimulated an axon twice, two EPSPs were recorded.
- If the delay between EPSPs was short enough, the second EPSP added to what was left of the first one, producing temporal summation.

The Properties of Synapses

c) Spatial Summation

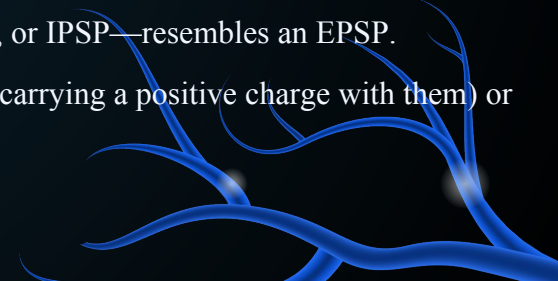
- Sherrington also found that synapses have the property of **spatial summation**—that is, summation over space. **Synaptic inputs from separate locations combine their effects on a neuron.**
- Sherrington again began with a pinch too weak to elicit a reflex. This time, instead of pinching one point twice, he pinched two points at once. Although neither pinch alone produced a reflex, together they did.
- Sherrington concluded that pinching two points activated separate sensory neurons, whose axons converged onto a neuron in the spinal cord.
- Excitation from either sensory axon excited that spinal neuron, but not enough to reach the threshold.
- *A combination of excitations exceeded the threshold and produced an action potential*
- Spatial summation is critical to brain functioning. Sensory input to the brain arrives at synapses that individually produce weak effects.
- However, each neuron receives many incoming axons that frequently produce synchronized responses
- Spatial summation ensures that a sensory stimulus stimulates neurons enough to activate them.
- Temporal summation and spatial summation ordinarily occur together. That is, a neuron might receive input from several axons at approximately, but not exactly, the same time.





The Properties of Synapses

d) Inhibitory Synapses

- When Sherrington vigorously pinched a dog's foot, the flexor muscles of that leg contracted, and so did the extensor muscles of the other three legs
 - Also, the dog relaxed the extensor muscles of the stimulated leg and the flexor muscles of the other legs.
 - Sherrington's explanation assumed certain connections in the spinal cord: A pinch on the foot sends a message along a sensory neuron to an interneuron (an intermediate neuron) in the spinal cord, which in turn excites the motor neurons connected to the flexor muscles of that leg
 - Sherrington surmised that the interneuron also sends a message to block activity of motor neurons to the extensor muscles in the same leg and the flexor muscles of the three other legs.
 - Later researchers physiologically demonstrated the inhibitory synapses that Sherrington had inferred. At these synapses, input from an axon hyperpolarizes the postsynaptic cell.
 - That is, it increases the negative charge within the cell, moving it further from the threshold and decreasing the probability of an action potential.
 - This **temporary hyperpolarization of a membrane—called an inhibitory postsynaptic potential**, or IPSP—resembles an EPSP.
 - An IPSP occurs when synaptic input selectively opens the gates for potassium ions to leave the cell (carrying a positive charge with them) or for chloride ions to enter the cell (carrying a negative charge).
- 

Relationship Among EPSP, IPSP, and Action Potentials

□ Sherrington's work opened the way to exploring the wiring diagram of the nervous system because **how neurons are connected determines how the brain and body respond**. Neurons can produce different responses depending on:

- ✓ Which neurons connect with each other
- ✓ Which connections are excitatory (EPSPs)
- ✓ Which connections are inhibitory (IPSPs)
- ✓ How many synapses are involved

□ Figure 3.8 □ When neuron 1 excites neuron 3, it also excites neuron 2, which inhibits neuron 3. The excitatory message reaches neuron 3 faster because it goes through just one synapse instead of two. The result is brief excitation (EPSP) in neuron 3, which stops quickly. You see how the inhibitory neurons, which are typically very small, can regulate the timing of activity.

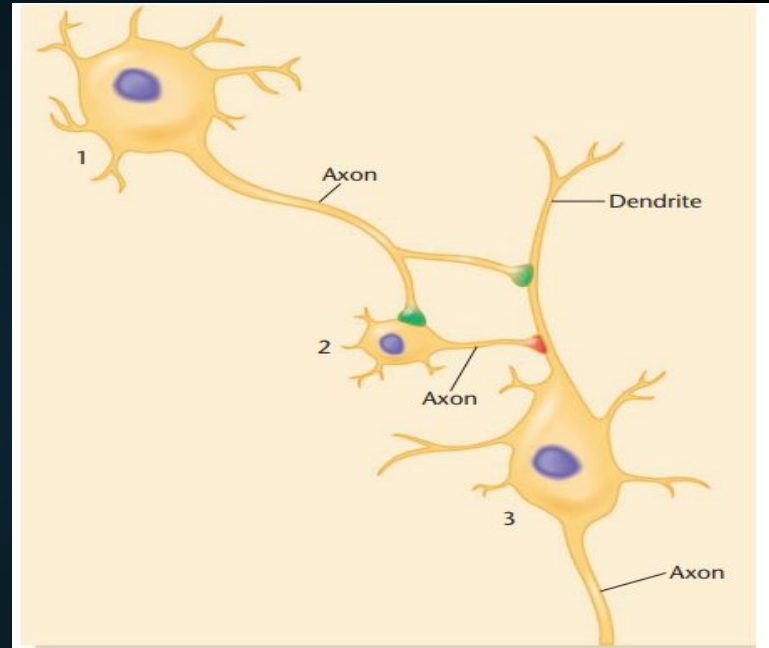


FIGURE 3.8 One of many possible wiring diagrams for synapses

Excitatory synapses are in green, and inhibitory synapses in red. In the circuit shown here, excitation would reach the dendrite before inhibition. (Remember, any transmission through a synapse produces a delay.) Therefore, the result would be brief excitation of the dendrite and then a stop. Inhibitory synapses serve many other functions, too. (Based on Kullmann & Lamsa, 2007)

Relationship Among EPSP, IPSP, and Action Potentials (Cont.)

- Neurons continuously receive many excitatory and inhibitory inputs. The way these inputs are connected and combined determines whether the neuron fires and how frequently it produces action potentials.
- Figure 3.9 □ either the axon from cell A or the axon from cell B stimulates cell X enough to reach its threshold. Therefore cell X responds to “A or B.”
- Figure 3.10 □ neither A nor B stimulates cell X enough to reach its threshold, but the two can summate to reach the threshold (spatial summation.) Therefore in this case cell X responds to “A and B.”
- Figure 3.11 □ cell X responds to “A and B if not C.”
- Sherrington assumed that synapses simply produce on and off responses. In fact, synapses vary enormously in their duration of effects.
- The effect of two synapses at the same time can be more than double the effect of either one, or less than double
- Most neurons have a **spontaneous firing rate**, a periodic production of action potentials even without synaptic input.
- In such cases, the EPSPs increase the frequency of action potentials above the spontaneous rate, whereas IPSPs decrease it.
- For example, if the neuron's spontaneous firing rate is 10 action potentials per second, a stream of EPSPs might increase the rate to 15 or more, whereas a preponderance of IPSPs might decrease it to 5 or fewer.

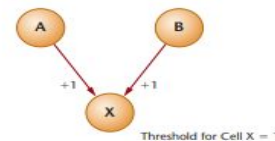


FIGURE 3.9 Wiring diagram for an “A or B” response
The axon from either A or B stimulates cell X enough to reach its threshold. (© Cengage Learning 2013)

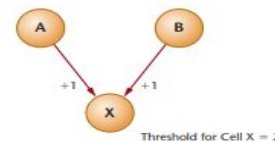


FIGURE 3.10 Wiring diagram for an “A and B” response
The axons from A and B stimulate cell X but neither one by itself reaches the threshold for X. The combination of both at the same time reaches the threshold. (© Cengage Learning 2013)

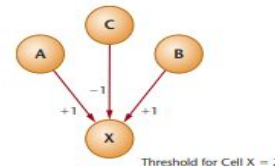
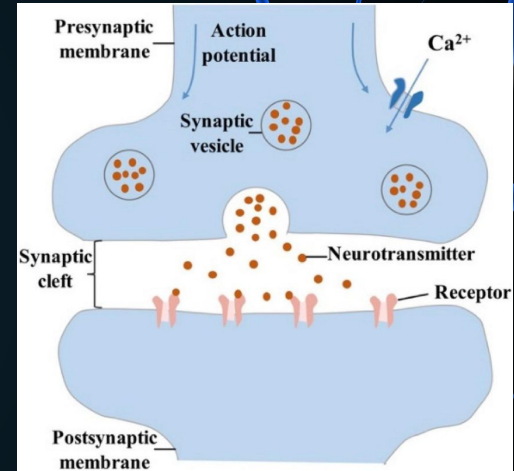


FIGURE 3.11 Wiring diagram for an “A and B if not C” response
The axons from A and B can combine to reach the threshold for X, but the axon from C can inhibit X enough to prevent a response. (© Cengage Learning 2013)

Chemical Events at the Synapse

- ❑ Although Charles Sherrington accurately inferred many properties of the synapse, he was wrong about one important point:
- ❑ Sherrington thought the slight delay at the synapse was too short for chemicals to be involved, so he concluded that synaptic transmission must be electrical.
- ❑ We now know that most synapses are chemical, using neurotransmitters, and these chemical synapses are very fast and versatile, producing either excitatory or inhibitory effects.



The Discovery of Chemical Transmission at Synapses

How did scientists find out that synapses can transmit signals chemically?

- EXPERIMENTS WITH HEART FUNCTION - A set of nerves called the sympathetic nervous system accelerates the heartbeat, relaxes the stomach muscles, dilates the pupils of the eyes, and regulates other organs.
- 1. **T. R. Elliott**, a British scientist reported in **1905** that applying the hormone **adrenaline** directly to the surface of the heart, the stomach, and the pupils produces the same effects as those of the sympathetic nervous system.
 - Elliott therefore suggested that the sympathetic nerves stimulate muscles by releasing adrenaline or a similar chemical.
 - However, Elliott's evidence was not convincing because he did **not prove yet that sympathetic nerves actually release adrenaline**. One possibility might be that adrenaline might simply **mimic** the nerve's effects.
- 2. **Otto Loewi**, a German physiologist, in **1920**, repeatedly stimulated a **frog's vagus nerve**, thereby decreasing the heart rate.
 - He then collected fluid from that heart, transferred it to a second frog's heart, and found that the second heart also decreased its rate of beating
 - Then Loewi stimulated the **accelerator nerve** to the first frog's heart, increasing the heart rate.
 - When he collected fluid from that heart and transferred it to the second frog's heart, its heart rate increased.
 - He knew he was collecting and transferring chemicals, not loose electricity.
 - Therefore, Loewi concluded, nerve stimulation can cause the release of a chemical that carries the message.
- 3. In **1950s** □ researchers established that **chemical transmission predominates throughout the nervous system.**

The Sequence of Chemical Events at a Synapse

1

The neuron synthesizes chemicals that serve as neurotransmitters. It synthesizes the smaller neurotransmitters in the axon terminals and synthesizes neuropeptides in the cell body.

2

Action potentials travel down the axon. At the presynaptic terminal, an action potential enables calcium to enter the cell. Calcium releases neurotransmitters from the terminals and into the synaptic cleft, the space between the presynaptic and postsynaptic neurons.

3

The released molecules diffuse across the cleft, attach to receptors, and alter the activity of the postsynaptic neuron.

4

The neurotransmitter molecules separate from their receptors.

5

The neurotransmitter molecules may be taken back into the presynaptic neuron for recycling or they may diffuse away

6

Some postsynaptic cells send reverse messages to control the further release of neurotransmitter by presynaptic cells.

The Sequence of Chemical Events at a Synapse

a) Types of Neurotransmitters

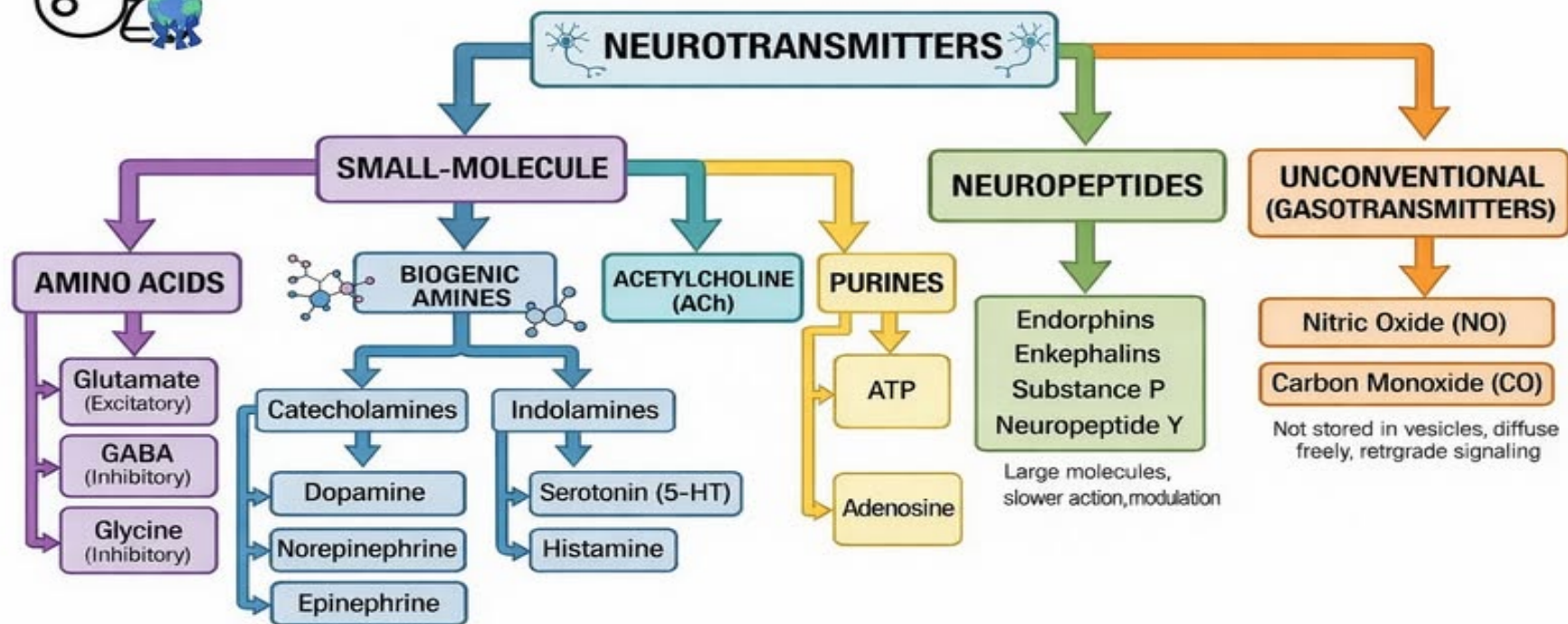
What are neurotransmitters?

□ At a synapse, a neuron releases chemicals that affect another neuron. Those chemicals are known as **neurotransmitters**

<u>S.No</u>	<u>Category</u>	<u>Description of Category</u>	<u>Examples</u>
1	Amino acids	Acids containing an amine group (NH ₂)	Glutamate, GABA, glycine, aspartate, etc
2	Monoamines	Chemicals formed by a change in certain amino acids	Acetylcholine
3	Acetylcholine	A chemical similar to an amino acid, except that it includes an N(CH ₃) ₃ group instead of an NH ₂	<u>Indoleamines</u> : serotonin <u>Catecholamines</u> : dopamine, norepinephrine, epinephrine
4	Neuropeptides	Chains of amino acids used by neurons as a neurotransmitter.	Endorphins, substance P, neuropeptide Y, etc
5	Purines	A category of chemicals including adenosine and several of its derivatives	ATP, adenosine, etc
6	Gases	Nitric oxide and possibly others	NO (nitric oxide), etc



CLASSIFICATION OF NEUROTRANSMITTERS



b) Synthesis of Transmitters

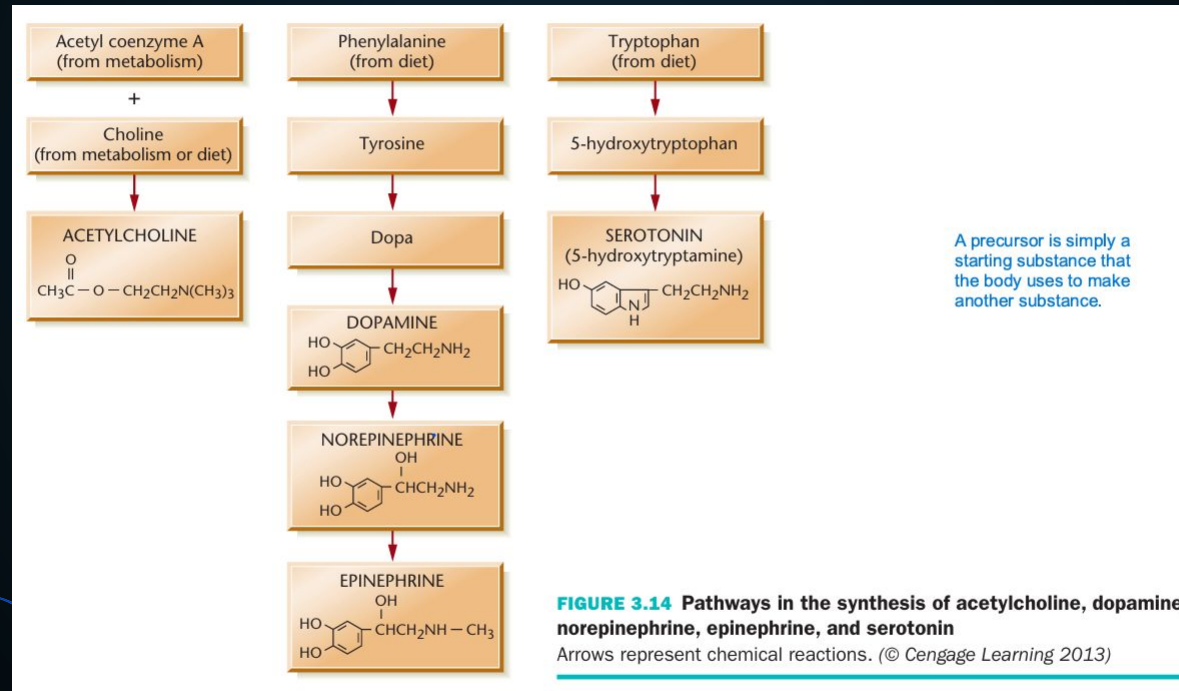
- Neurons synthesize nearly all neurotransmitters from amino acids, which the body obtains from proteins in the diet.
- Epinephrine, norepinephrine, and dopamine fall under the category known as **catecholamines**, because they contain a catechol group and an amine group

How are neurotransmitters synthesized?

- **Acetylcholine** is synthesized from choline, which is abundant in milk, eggs, and peanuts.
- The amino acids phenylalanine and tyrosine, present in protein rich foods like, are precursors of **dopamine, norepinephrine, and epinephrine.**
- People with phenylketonuria lack the enzyme ? that converts phenylalanine to tyrosine. They can get tyrosine from their diet, but they need to minimize intake of phenylalanine.
- The amino acid tryptophan, the precursor to **serotonin**, crosses the blood–brain barrier by a special transport system that it shares with other large amino acids.
- The amount of tryptophan in the diet controls the amount of serotonin in the brain, so your serotonin levels rise after you eat foods richer in tryptophan, such as soy, and fall after something low in tryptophan, such as maize (American corn).
- However, tryptophan has to compete with other, more abundant large amino acids, such as phenylalanine, that share the same transport system.

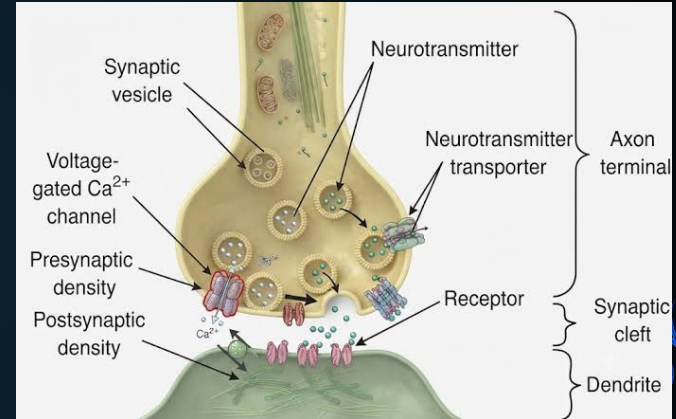
b) Synthesis of Transmitters (Cont.)

- One way to increase tryptophan entry to the brain is to decrease consumption of phenylalanine.
- Another is to eat carbohydrates. Carbohydrates increase the release of the hormone insulin, which takes several competing amino acids out of the bloodstream and into body cells, thus decreasing the competition against tryptophan.



c) Storage of Neurotransmitters

- Most neurotransmitters are synthesized in the presynaptic terminal, near the point of release.
- The **presynaptic terminal stores high concentrations of neurotransmitter molecules in vesicles**, tiny nearly spherical packets
- The presynaptic terminal also maintains many neurotransmitter molecules **outside the vesicles**.
- It is possible for a neuron to accumulate excess levels of a neurotransmitter. Thus, neurons that release serotonin, dopamine, or norepinephrine contain an enzyme, **MAO (monoamine oxidase)**, that breaks down these transmitters into inactive chemicals.



d) Release and Diffusion of Transmitters

- At the end of an axon, the action potential itself does not release the neurotransmitter.
- Rather, the **depolarization opens voltage-dependent calcium gates in the presynaptic terminal**.
- Within 1 or 2 milliseconds (ms) after **calcium enters the presynaptic terminal**, it causes **exocytosis—release of neurotransmitter in bursts from the presynaptic neuron into the synaptic cleft** that separates one neuron from another.
- After its release from the presynaptic cell, the **neurotransmitter diffuses across the synaptic cleft to the postsynaptic membrane**, where it **attaches to a receptor**.
- The neurotransmitter takes no more than 0.01 ms to diffuse across the cleft, which is only 20 to 30 nanometers (nm) wide.
- Sherrington did not believe chemical processes could be fast enough to account for the activity at synapses. He did not imagine such a narrow gap through which chemicals could diffuse so quickly.
- For many years, investigators believed that each neuron released just one neurotransmitter, but later researchers found that many release a combination of two or more transmitters

Why does a neuron release a combination of transmitters instead of just one?

- The combination makes the neuron's message more complex, such as brief excitation followed by slight but prolonged inhibition. Although a neuron releases only a limited number of neurotransmitters, it may receive and respond to many neurotransmitters at different synapses. For example, at various locations on its membrane, it might have receptors for glutamate, serotonin, acetylcholine, and others.

e) Activating Receptors of the Postsynaptic Cell

- Sherrington's concept of the synapse was simple: Input produced excitation or inhibition—i.e., on/off.
- When Eccles recorded from individual cells that cells produced only brief EPSPs and IPSPs— again, just on/off.
- Researchers discovered more and more neurotransmitters and wondered, “Why does the nervous system use so many chemicals, if they all produce the same type of message (either EPSP or IPSP)?”
- The effect of a neurotransmitter depends on its receptor on the postsynaptic cell. When the neurotransmitter attaches to its receptor, the receptor may open a channel— exerting an ionotropic effect—or it may produce a slower but longer effect—a metabotropic effect.

e) Activating Receptors of the Postsynaptic Cell

IONOTROPIC EFFECTS

- When the **neurotransmitter binds to an ionotropic receptor**, it **twists the receptor** enough to **open its central channel**, which is shaped to let a particular type of ion pass through.
- In contrast to the sodium and potassium channels along an axon, which are voltage-gated, the channels controlled by a neurotransmitter are **transmitter-gated or ligand-gated channels**. (A ligand is a chemical that binds to another chemical.) That is, when the neurotransmitter attaches, it opens a channel.
- Ionotropic effects **begin quickly, sometimes within less than a millisecond** after the transmitter attaches. The effects decay with a half-life of about 5 ms.
- They are **well suited to conveying visual information, auditory information**, and anything else that needs to be updated as quickly as possible.
- Most of the brain's **excitatory ionotropic synapses** use the neurotransmitter glutamate.
- Most of the **inhibitory ionotropic synapses** use the neurotransmitter GABA (gamma-amino-butyric acid), which opens chloride gates, enabling chloride ions, with their negative charge, to cross the membrane into the cell more rapidly than usual.
- Glycine is another common inhibitory transmitter, found mostly in the spinal cord
- Acetylcholine, another transmitter at many ionotropic synapses, is excitatory in most cases.

e) Activating Receptors of the Postsynaptic Cell

METABOTROPIC EFFECTS AND SECOND MESSENGER SYSTEMS

- Neurotransmitters also exert metabotropic effects by initiating a sequence of metabolic reactions that are slower and longer lasting than ionotropic effects
- Metabotropic effects emerge 30 ms or more after the release of the transmitter. Typically, they last up to a few seconds, but sometimes longer.
- Metabotropic synapses use many neurotransmitters like dopamine, norepinephrine, and serotonin and some times glutamate and GABA too.
- When a neurotransmitter attaches to a metabotropic receptor, it bends the receptor protein that goes through the membrane of the cell. The other side of that receptor is attached to a G protein—that is, a protein coupled to guanosine triphosphate (GTP), an energy storing molecule. Bending the receptor protein detaches that G protein, which is then free to take its energy elsewhere in the cell
- The result of that G protein is increased concentration of a second messenger, such as cyclic adenosine monophosphate (cyclic AMP), inside the cell.
- Just as the “first messenger” (the neurotransmitter) carries information to the postsynaptic cell, the second messenger communicates to many areas within the cell.
- It may open or close ion channels in the membrane or activate a portion of a chromosome
- Metabotropic synapses are better suited for more enduring effects such as taste, smell, pain, arousal, attention, pleasure, and emotion.

e) Activating Receptors of the Postsynaptic Cell

NEUROPEPTIDES

Researchers often refer to neuropeptides as neuromodulators, because they have several properties that set them apart from other transmitters

TABLE 3.2 Distinctive Features of Neuropeptides

	Neuropeptides	Other Neurotransmitters
Place synthesized	Cell body	Presynaptic terminal
Place released	Mostly from dendrites, also cell body and sides of axon	Axon terminal
Released by	Repeated depolarization	Single action potential
Effect on neighboring cells	They release the neuropeptide too	No effect on neighbors
Spread of effects	Diffuse to wide area	Effect mostly on receptors of the adjacent postsynaptic cell
Duration of effects	Many minutes	Generally less than a second to a few seconds

Neuropeptides

- ❖ Neuropeptides are small protein like molecules (peptides) used by neurons to communicate with each other. (Autocrine/paracrine).
- ❖ Neuronal signaling molecules(not recycled back into the cell once secreted, unlike glutamate, dopamine, serotonin etc.)
- ❖ Responsible for brain function:
 - Analgesia.
 - Food intake
 - Learning & memory.
 - Metabolism, reproduction.
 - Social behaviors.

Eg: Neuropeptide Y(NPY), Cholecystokinin(CCK), Tachykinins(substance P, Neurokinin), Arginine Vasopressin(AVP), Corticotropin releasing factor(CRF)

e) Activating Receptors of the Postsynaptic Cell

HORMONES

- **Hormone** □ a chemical that is secreted by cells in one part of the body and conveyed by the blood to influence other cells.
- **Difference** □ a neurotransmitter is like a telephone signal: It conveys a message from the sender to the intended receiver. Hormones function more like a radio station: They convey a message to any receiver tuned in to the right station. Neuropeptides are intermediate. They are like hormones, except that they diffuse only within the brain, and the blood doesn't carry them to other parts of the body.
- Hormones are particularly useful for coordinating slow but long lasting changes in multiple parts of the body.
- For example, birds that are preparing for migration secrete hormones that change their eating and digestion to store extra energy for a long journey.
- Among the various types of hormones are **protein hormones** and **peptide hormones**, composed of chains of amino acids. (Proteins are longer chains and peptides are shorter.)
- Protein and peptide hormones attach to membrane receptors, where they activate a second messenger within the cell—exactly like a **metabotropic synapse**.

e) Activating Receptors of the Postsynaptic Cell

HORMONES (Cont.)

- In fact, many chemicals serve as both neurotransmitters and hormones. Just as circulating hormones modify brain activity, hormones secreted by the brain control the secretion of many other hormones.
- Pituitary gland, attached to the hypothalamus consists of two distinct glands, the anterior pituitary and the posterior pituitary, which release different sets of hormones
- Neurons in the hypothalamus synthesize the hormones oxytocin and vasopressin (also known as antidiuretic hormone), which migrate down axons to the posterior pituitary
- Later, the posterior pituitary releases these hormones into the blood.
- The anterior pituitary, composed of glandular tissue, synthesizes six hormones, although the hypothalamus controls their release
- The hypothalamus secretes releasing hormones, which flow through the blood to the anterior pituitary.
- There they stimulate or inhibit the release of the following hormones: The hypothalamus maintains fairly constant circulating levels of certain hormones through a negative feedback system. For example, when the level of thyroid hormone is low, the hypothalamus releases TSH-releasing hormone, which stimulates the anterior pituitary to release TSH, which in turn causes the thyroid gland to secrete more thyroid hormones

Hormone VS Neurotransmitter

Comparison Chart

Characteristics	Hormone	Neurotransmitter
Organ System	Endocrine System	Nervous System
Mode of Transmission	Through blood stream	Across synaptic cleft
Transmission Speed	May take a few minutes or a few days	Usually within milliseconds
Transmission Distance	Act on a distant site from where it is produced	Reacts in direct proximity to their target cells
Function	They affect physiological processes such as growth and development, metabolism, mood, sexual function, reproduction, etc.	They facilitate transmission between neurons by passing action potential from the axon of one neuron to the dendrite of the next neuron
Types	Amino acid-based hormones and steroids	-According to Facilitation of Ion Flow: excitatory and inhibitory -According to structure: Small molecule and neuropeptides
Unique Capability	Regulation of the target organs and tissues	Only capable of stimulating postsynaptic neuron

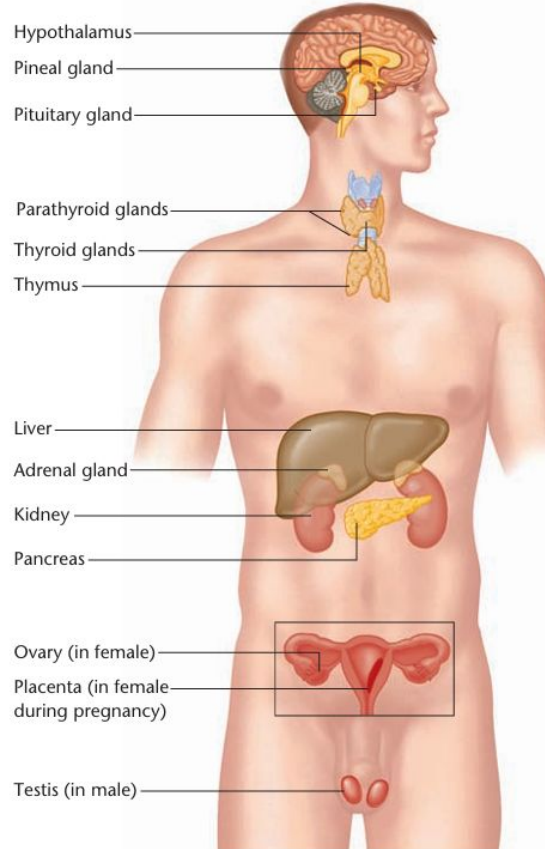


FIGURE 3.18 Location of some major endocrine glands
(Source: Starr & Taggart, 1989)

f) Inactivation and Reuptake of Neurotransmitters

- A neurotransmitter does not linger at the postsynaptic membrane.
- If it did, it might continue exciting or inhibiting the receptor.

Various neurotransmitters are inactivated in different ways.

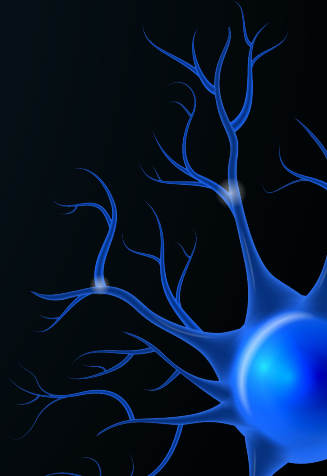
1. After acetylcholine activates a receptor, it is **broken down by the enzyme** acetylcholinesterase into two fragments: acetate and choline.
 - The choline diffuses back to the presynaptic neuron, which takes it up and reconnects it with acetate already in the cell to form acetylcholine again.
2. On the other hand, serotonin and the catecholamines do not break down into inactive fragments at the postsynaptic membrane.
 - They **simply detach from the receptor and the presynaptic neuron takes up most of the released neurotransmitter molecules intact and reuses them**. This process, called **reuptake**, occurs through special membrane proteins called **transporters**.

f) Inactivation and Reuptake of Neurotransmitters

- Individuals vary in the activity of certain transporters.
- Genetic variations in serotonin transporters relate to individual differences in anxiety
- Transporters also differ in their abundance from one brain area to another.
- Dopamine transporters in the caudate nucleus are highly efficient, and reuptake accounts for nearly all of the released dopamine. In other brain areas, fewer transporters are present, and reuptake is slower. If dopamine is released rapidly in those areas, it accumulates and an enzyme called COMT (catechol-o methyltransferase) breaks down the excess into inactive chemicals that cannot stimulate the dopamine receptors. Those breakdown products wash away and eventually show up in the blood and urine. In the prefrontal cortex, COMT breaks down about half of the released dopamine (Yavich, Forsberg, Karayiorgou, Gogos, & Männistö, 2007). A consequence is that neurons in that area easily diminish their supply of dopamine, and they cannot release dopamine rapidly for long. The neuropeptides are neither inactivated nor reabsorbed. They simply diffuse away. Because these large molecules are resynthesized slowly, a neuron can temporarily exhaust its supply.



g) Negative Feedback from the Postsynaptic Cell

- Many presynaptic terminals have receptors sensitive to the same transmitter they release.
 - These receptors are known as **autoreceptors**—receptors that respond to the released transmitter by inhibiting further synthesis and release.
 - That is, they provide negative feedback
 - Also, some postsynaptic neurons respond to stimulation by releasing special chemicals that travel back to the presynaptic terminal, where they inhibit further release of the transmitter - **retrograde signal**
 - Nitric oxide is one such transmitter.
 - Two others are anandamide and 2-AG (sn-2 arachidonylglycerol). Thus, postsynaptic neurons have ways to control or limit their own input.
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h) Electrical Synapses

- A few special-purpose synapses operate electrically.
- Electrical transmission is faster than even the fastest chemical transmission, and electrical synapses have evolved in cases where exact synchrony between two cells is important.
- For example, some of the cells that control your rhythmic breathing are synchronized by electrical synapses. (It's important to inhale on the left side at the same time as on the right side.)
- At an electrical synapse, the membrane of one neuron comes into direct contact with the membrane of another, as shown in Figure 3.22.
- This contact is called a **gap junction**.
- Fairly large pores of the membrane of one neuron line up precisely with similar pores in the membrane of the other cell.
- These pores are large enough for sodium and other ions to pass readily, and unlike the other membrane channels we have considered, these pores remain open constantly.
- Therefore, whenever one of the neurons is depolarized, sodium ions from that cell can pass quickly into the other neuron and depolarize it, too.
- As a result, the two neurons act almost as if they were in fact a single neuron.

THANK YOU

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